Zahra TehraniNasab

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zahra-tehraninasab

Education

McGill University — **Montreal, Canada**
M.Sc. Electrical Engineering (Thesis), GPA: 3.93/4 — *Aug 2023 – Present*

Sharif University of Technology — **Tehran, Iran**
B.Sc. Computer Science, GPA: 17.64/20 — *Sep 2018 – Feb 2023*

National Organization for Development of Exceptional Talents (Sampad) — **Shiraz, Iran**
High School Diploma, GPA: 19.99/20 — *Sep 2015 – June 2018*

Honors & Awards

Awarded the Mila EDI Scholarships - Women in AI Category, Valued at CAD$10,000, MILA (Quebec AI Institute), 2024-2025

Awarded the Graduate Excellence Fellowship (GEF), Valued at CAD$4,500, McGill University, 2023

Third Prize, MICCAI Educational Challenge (2025), MICCAI Educational Challenge for a blog post on generative models in medical imaging.

Ranked 229th Among More Than 140,000 (Top 0.17%) Participants, Iranian Nationwide University Entrance Exam, Mathematics and Physics Discipline, 2018

Publications

PIKACHU: Prototypical In-context Knowledge Adaptation for Clinical Heterogeneous Usage, MIDL 2026, Amar Kumar, **Zahra TehraniNasab**, Emily Kaczmarek, Tal Arbel

Pixel Perfect MegaMed: A Megapixel-Scale Vision-Language Foundation Model for Generating High Resolution Medical Images, MICCAI (2025) DGM4MICCAI Workshop Proceedings, **Zahra TehraniNasab**, Hujun Ni, Amar Kumar, Tal Arbel, *Project Website*

Language-Guided Trajectory Traversal in Disentangled Stable Diffusion Latent Space for Factorized Medical Image Generation, CVPR (2025) MIV Workshop Proceedings, **Zahra TehraniNasab***, Amar Kumar*, Tal Arbel, *Project Website*

Pixels Under Pressure: Exploring Fine-Tuning Paradigms for Foundation Models in High-Resolution Medical Imaging, MICCAI (2025) ELAMI Workshop Proceedings, **Zahra TehraniNasab**, Amar Kumar, Tal Arbel, *Arxiv Link*

RL4Med-DDPO: Reinforcement Learning for Controlled Guidance Towards Diverse Medical Image Generation using Vision-Language Foundation Models, MICCAI (2025) Main Track, Parham Saremi*, Amar Kumar*, Mohammed Mohammed, **Zahra TehraniNasab**, Tal Arbel, *Project Website*

Discovering Latent Knowledge Graphs for Capturing Diversity in Conditional Image Generation. NeurIPS (2025) Main Track, Bailey Trang, Parham Saremi, Alan Q. Wang, Fangrui Huang, **Zahra TehraniNasab**, Amar Kumar, Tal Arbel, Li Fei-Fei, Ehsan Adeli *Arxiv Link*

Towards Reliable Human Pose Forecasting with Uncertainty, IEEE Robotics and Automation Letters, S. Saadatnejad, **Z. TehraniNasab***, M. Mirmohammadi*, M. Daghyani*, P. Saremi*, Y. Zoroofchi Benisi*, A. Alimohammadi*, T. Mordan, A. Alahi, *Arxiv Link*

Experience

Industry Research

Research Intern, Mila - Quebec AI Institute — **Montreal, Canada**
Applied Machine Learning Team (ALMRT) — *Oct 2025 – Feb 2026*
- Developed a **multimodal learning** framework for prospectivity mapping for geospatial mineral exploration.

Academic Research

Graduate Student Researcher, McGill and MILA - Quebec AI Institute — **Montreal, Canada**
Generative Modeling, Explainability, Fairness, Synthetic Data Generation — *Sep 2023 – Present*
- Developed **vision–language foundation models (VLMs)** to improve generalization in healthcare

applications on a **large-scale dataset** of medical images.
- **Scaled** diffusion-based medical image synthesis to ultra-high resolutions (1024×1024 and 2048×2048).
- Mitigated the computational demands of **large-scale vision–language foundation models** through **parameter-efficient fine-tuning (LoRA/adapters)** and **distributed multi-GPU/multi-node** training.
- Built **reproducible and reliable ML training and evaluation pipelines** to support **large-scale experimentation**.
- Investigated and characterized latent subspace structure in **diffusion-based vision–language models** to enable interpretable and controllable generation.
- **Augmented** the training data via **diffusion-based synthesis** and retrained the model to **improve robustness and generalization**.

Research Intern, École polytechnique fédérale de Lausanne (EPFL) — **Remote Internship**
Human Pose Prediction, VITA Lab — *Dec 2021 – Jul 2022*
- Improved state-of-the-art pose prediction models by integrating **Predictive Uncertainty Awareness**.
- Collaborated with five colleagues on developing and maintaining a general **deep learning library** for **Human Pose Prediction** using Python and PyTorch.
- Developed pose prediction models leveraging **transformer-based models** for **sequence modeling** of **human motion**.

Research Intern, University of California Irvine (UCI) — **Remote Internship**
Food Computing — *Dec 2021 – Jul 2022*
- Joint research project between the the **University of California Irvine (UCI) and the Sharif University of Technology (SUT)**
- Developed and experimented with a hierarchical visual feature extractor to predict ingredients and categories of food from an image.
- Developed and experimented with a Graph-LSTM network to predict food categories from the food recipe graph.

Teaching Assistant

Applied Machine Learning, McGill University — *Jan 2025 – April 2025, Sep 2025 – Dec 2025*
- Held tutorials on machine learning related concepts

Fundamentals of Machine Learning, McGill University — *Aug 2024 – Dec 2024*
- Held a PyTorch tutorial for students
- Designed a Kaggle competition as an assignment for students

Artificial Intelligence, Sharif University of Technology — *Feb 2021 – Jul 2022, Sep 2022 – Dec 2022*
- Designed supplementary material on Transformer models for students.
- Design and grade practical assignments for the course's deep learning and machine learning chapter.

Computer Systems, Sharif University of Technology — *Sep 2021 - Dec 2021*
- Designed and graded questions for two assignments of the course.

Networking, Sharif University of Technology — *Feb 2021 – Jul 2021, Feb 2022 – Jul 2022*
- Designed and graded questions for the course's assignment and midterm exam.

Relevant Courses

CIFAR DLRL Summer School (2025) — *Certificate*
Statistical Computer Vision, McGill University — *Grade: 4/4*
Natural Language Processing, McGill University — *Grade: 4/4*
Machine Learning In Network Biology, McGill University — *Grade: 4/4*
Representation Learning, University of Montreal
B.Sc. Thesis on Virtual Try-on, Sharif University of Technology — *Grade: 20/20*
- Investigated and wrote a review on state-of-the-art virtual-try on papers, including the following concepts: Conditional Generative Adversarial Networks (C-GANs), **Knowledge Distillation** (Teacher-Student Networks), Human Pose Estimation, Inpainting, Optical Flow

Deep Learning, Neuromatch Academy - International Summer School — *Certificate*
Computer Vision, Sharif University of Technology — *Grade: 19.3/20*
Natural Language Processing, Sharif University of Technology — *Grade: 20/20*
Image Processing, Sharif University of Technology — *Grade: 19.7/20*
Machine Learning, Sharif University of Technology — *Grade: 18.8/20*
Neural Networks and Deep Learning, Coursera - Online Course — *Certificate*

Skills

Programming languages: Python | Typescript | F# | Java | Matlab | Octave
Operating systems and developer tools: Linux | Git | GitHub | Tmux
Deep Learning skills and tools: Vision-Language Foundation Models (VLMs), Large Language Models (LLMs), Knowledge Retrieval, Generative Modeling, Computer Vision, Natural Language Processing, Sequence Modeling, Working with GPU clusters, Working with distributed processing on multiple GPUs and Devices, Pytorch, Numpy, Pandas, Sklearn, Matplotlib, HuggingFace, CometML